

ELITE IGCSE ACADEMY

Student Past Paper Solutions

May-Nov 2020 P1HR Worked Solutions

MayNov 2020 | P1HR

27 questions ■ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION**Q#: 1****Marks: 3****TOPIC: Fractions**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

1 Show that $3\frac{3}{4} \times \frac{7}{9} = 2\frac{11}{12}$

(Total for Question 1 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 1** **Marks: 3****TOPIC: Fractions**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Simplify fraction**

$$3\frac{3}{4} = \frac{15}{4}$$

2. Simplify fraction

Now multiply:

$$\frac{15}{4} \times \frac{7}{9}$$

3. Simplify fraction

Cancel 15 and 9:

$$\frac{5}{4} \times \frac{7}{3} = \frac{35}{12}$$

$$\frac{35}{12} = 2\frac{11}{12}$$

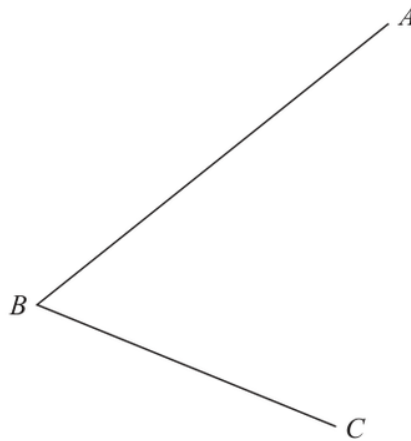
FINAL ANSWER

$$2\frac{11}{12}$$

PAST PAPER QUESTION**Q#: 2****Marks: 2****TOPIC: Bearings, Scale Drawing & Constructions**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

- 2 Using ruler and compasses only, construct the bisector of angle ABC .
You must show all your construction lines.



(Total for Question 2 is 2 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 2

TOPIC: Bearings, Scale Drawing & Constructions

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Construction answer

To construct the bisector of angle ABC :

1. Put the compass point at B and draw an arc cutting BA and BC .
2. From those two cut points, draw equal arcs inside the angle.
3. Draw a straight line from B through the point where the two arcs cross.

FINAL ANSWER

This line bisects angle ABC .

EA

PAST PAPER QUESTION**Q#: 3****Marks: 6****TOPIC: Solving Linear Equations**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

3 (a) Simplify $h^7 \times h^2$
(1)

$$G = c^2 - 4c$$

(b) Find the value of G when $c = -5$ $G =$
(2)(c) Solve $\frac{5x-3}{4} = 2x+3$

Show clear algebraic working.

 $x =$
(3)**(Total for Question 3 is 6 marks)**

PAST PAPER SOLUTION**Q#: 3****Marks: 6****TOPIC: Solving Linear Equations**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$h^7 \times h^2 = h^{7+2} = h^9$$

2. (b)

(b)

$$G = c^2 - 4c$$

3. Solve quadratic equationWhen $c = -5$:

$$G = (-5)^2 - 4(-5) = 25 + 20 = 45$$

4. (c)

(c)

$$\frac{5x - 3}{4} = 2x + 3$$

5. Multiply by 4

Multiply by 4:

$$5x - 3 = 8x + 12$$

$$-3x = 15$$

$$x = -5$$

FINAL ANSWER h^9 , $G = 45$, and $x = -5$.

PAST PAPER QUESTION**Q#: 4****Marks: 5****TOPIC: Statistics Toolkit**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

- 4 The table gives information about the length of time, in minutes, that each of 60 students took to travel to school on Monday.

Length of time (t minutes)	Frequency
$0 < t \leq 10$	4
$10 < t \leq 20$	10
$20 < t \leq 30$	15
$30 < t \leq 40$	25
$40 < t \leq 50$	6

- (a) Write down the modal class interval.

.....
(1)

- (b) Work out an estimate for the mean length of time taken by these 60 students to travel to school on Monday.
Give your answer correct to one decimal place.

..... minutes
(4)

(Total for Question 4 is 5 marks)

PAST PAPER SOLUTION**Q#: 4****Marks: 5****TOPIC: Statistics Toolkit**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. The modal class is the**

The modal class is the class with the greatest frequency:

$$30 < t \leq 40$$

2. Use midpoints

Use midpoints 5, 15, 25, 35, 45.

$$5(4) + 15(10) + 25(15) + 35(25) + 45(6) = 1690$$

$$\text{estimated mean} = \frac{1690}{60} = 28.166 \dots$$

FINAL ANSWER

modal class $30 < t \leq 40$, estimated mean 28.2 minutes.

PAST PAPER QUESTION**Q#: 5****Marks: 6****TOPIC: Percentages**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

- 5** In 2017, the population of a village was 7500
In 2019, the population of the village was 8265

(a) Work out the percentage increase in the population of the village from 2017 to 2019

..... %
(3)

In a sale, normal prices are reduced by 30%
The sale price of a T-shirt was 31.50 euros.

(b) Work out the normal price of the T-shirt.

..... euros
(3)

(Total for Question 5 is 6 marks)

PAST PAPER SOLUTION**Q#: 5****Marks: 6****TOPIC: Percentages**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. The population increase is**

The population increase is

$$8265 - 7500 = 765$$

2. Percentage increase

Percentage increase:

$$\frac{765}{7500} \times 100 = 10.2\%$$

3. The T shirt sale price

The T-shirt sale price is after a 30% reduction, so it is 70% of the normal price.

$$0.70 \times \text{normal price} = 31.50$$

$$\text{normal price} = 31.50 \div 0.70 = 45$$

FINAL ANSWER

(a) 10.2%, (b) 45 euros.

PAST PAPER QUESTION**Q#: 6** **Marks: 2****TOPIC: Coordinate Geometry**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

- 6 Point A has coordinates $(-3, 11)$
Point B has coordinates $(47, b)$
The midpoint of AB has coordinates $(a, -19)$
Find the value of a and the value of b .

 $a = \dots\dots\dots$ $b = \dots\dots\dots$

(Total for Question 6 is 2 marks)

EA

PAST PAPER SOLUTION**Q#: 6****Marks: 2****TOPIC: Coordinate Geometry**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate value**

$$A = (3, 11), \quad B = (47, b)$$

2. The midpoint isThe midpoint is $(a, 19)$.**3. Evaluate fraction**

For the x-coordinate:

$$a = \frac{3 + 47}{2} = 25$$

4. Evaluate fraction

For the y-coordinate:

$$19 = \frac{11 + b}{2}$$

$$38 = 11 + b$$

$$b = 27$$

FINAL ANSWER

$$a = 25, b = 27.$$

PAST PAPER QUESTION**Q#: 7****Marks: 4****TOPIC: Standard & Compound Units**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

- 7 Pedro drove from Toulouse to Montpellier in 2 hours 42 minutes.
He drove at an average speed of 90 km/hour.

Janine drove from Toulouse to Montpellier along the same route as Pedro.
The journey took her 3 hours.

Work out Janine's average speed for the journey.

..... km/hour

(Total for Question 7 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 7** **Marks: 4****TOPIC: Standard & Compound Units**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Pedro's time was**

Pedro's time was

$$2 \text{ h } 42 \text{ min} = 2.7 \text{ h}$$

2. Distance from Toulouse to Montpellier

Distance from Toulouse to Montpellier:

$$90 \times 2.7 = 243 \text{ km}$$

3. Calculate value

Janine took 3 hours, so her average speed was

$$243 \div 3 = 81$$

FINAL ANSWER

81 km/hour.

PAST PAPER QUESTION**Q#: 8****Marks: 3****TOPIC: Compound Interest & Depreciation**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

- 8** Harold bought an antique clock for £1200
The clock increased in value by 8% per year.

Find the value of the clock exactly 3 years after Harold bought the clock.
Give your answer correct to the nearest £.

£

(Total for Question 8 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 8****Marks: 3****TOPIC: Compound Interest & Depreciation**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. The multiplier for an increase**

The multiplier for an 8% increase is

$$1.08$$

2. Value after 3 years

Value after 3 years:

$$1200(1.08)^3 = 1511.6544$$

3. Use compound interest

Correct to the nearest pound,

$$1511.6544 \approx 1512$$

FINAL ANSWER

1512 pounds

PAST PAPER QUESTION**Q#: 9****Marks: 3****TOPIC: Standard & Compound Units**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

- 9** A box is put on a horizontal table.

The face of the box in contact with the table is a square of side 1.5 metres.

The pressure on the table due to the box is 34.8 newtons/m²

Work out the force exerted by the box on the table.

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

..... newtons

(Total for Question 9 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 9 Marks: 3

TOPIC: **Standard & Compound Units**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. The area of the square

The area of the square face is

$$1.5 \times 1.5 = 2.25 \text{ m}^2$$

2. Use

Use

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

3. Calculate area

So

$$\text{force} = \text{pressure} \times \text{area}$$

$$34.8 \times 2.25 = 78.3$$

FINAL ANSWER

78.3 newtons.

PAST PAPER QUESTION**Q#: 10****Marks: 5****TOPIC: Percentages**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

10 Alex makes 80 cakes to sell.

He makes chocolate cakes, lemon cakes and fruit cakes where

$$\begin{array}{ccccc} \text{number of} & : & \text{number of} & : & \text{number of} \\ \text{chocolate cakes} & & \text{lemon cakes} & & \text{fruit cakes} \end{array} = 3 : 2 : 5$$

Alex sells

all of the chocolate cakes

 $\frac{3}{4}$ of the lemon cakes $\frac{7}{8}$ of the fruit cakes

The profit he makes on each cake he sells is shown in the table.

Type of cake	Profit per cake he sells
chocolate	£2.00
lemon	£1.70
fruit	£2.40

Work out the total profit that Alex makes from the cakes he sells.

£.....

(Total for Question 10 is 5 marks)

PAST PAPER SOLUTION**Q#: 10****Marks: 5****TOPIC: Percentages**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. The ratio parts are**

The ratio parts are

$$3 + 2 + 5 = 10$$

2. Numbers of cakes made

Numbers of cakes made:

$$\text{chocolate} = 80 \times \frac{3}{10} = 24$$

$$\text{lemon} = 80 \times \frac{2}{10} = 16$$

$$\text{fruit} = 80 \times \frac{5}{10} = 40$$

3. Cakes sold

Cakes sold:

$$\text{chocolate} = 24, \quad \text{lemon} = \frac{3}{4} \times 16 = 12, \quad \text{fruit} = \frac{7}{8} \times 40 = 35$$

4. Total profit

Total profit:

$$24(2.00) + 12(1.70) + 35(2.40) = 48 + 20.40 + 84 = 152.40$$

FINAL ANSWER

152.40 pounds

PAST PAPER QUESTION**Q#: 11****Marks: 5****TOPIC: Cumulative Frequency Diagrams**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

11 The frequency table gives information about the ages of the 80 people in a train carriage.

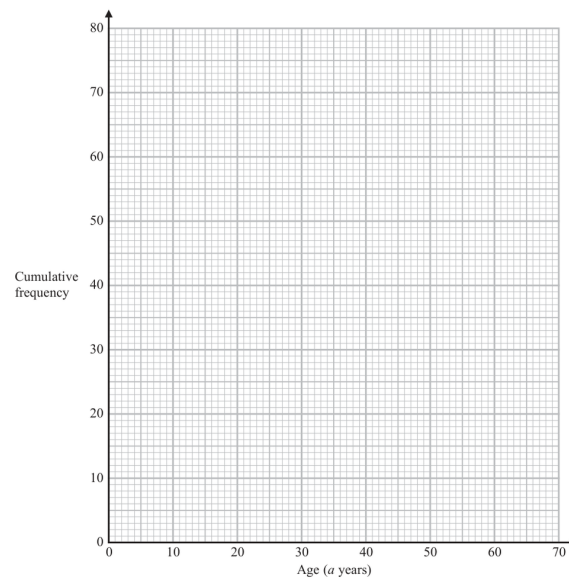
Age (a years)	Frequency
$0 < a \leq 20$	9
$20 < a \leq 30$	19
$30 < a \leq 40$	17
$40 < a \leq 50$	18
$50 < a \leq 60$	13
$60 < a \leq 70$	4

(a) Complete the cumulative frequency table.

Age (a years)	Cumulative frequency
$0 < a \leq 20$	
$0 < a \leq 30$	
$0 < a \leq 40$	
$0 < a \leq 50$	
$0 < a \leq 60$	
$0 < a \leq 70$	

(1)

(b) On the grid, draw a cumulative frequency graph for your table.



(2)

(c) Use your graph to find an estimate for the median age of the people in the train carriage.

_____ years
(2)

(Total for Question 11 is 5 marks)

PAST PAPER SOLUTION**Q#: 11****Marks: 5****TOPIC: Cumulative Frequency Diagrams**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. The cumulative frequencies are**

The cumulative frequencies are

9, 28, 45, 63, 76, 80

2. The median is the th

The median is the 40th value.

$$30 + \frac{40 - 28}{45 - 28} \times 10 \approx 37.1$$

FINAL ANSWER

median about 37 years.

PAST PAPER QUESTION**Q#: 12****Marks: 4****TOPIC: Simultaneous Equations**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

12 Solve the simultaneous equations

$$7x + 2y = 5.5$$

$$3x - 5y = 17$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 12 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 12****Marks: 4****TOPIC: Simultaneous Equations**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Solve simultaneous equations**

$$7x + 2y = 5.5$$

$$3x - 5y = 17$$

2. Multiply the first equation by

Multiply the first equation by 5:

$$35x + 10y = 27.5$$

3. Multiply the second equation by

Multiply the second equation by 2:

$$6x - 10y = 34$$

4. Add

Add:

$$41x = 61.5$$

$$x = 1.5$$

5. Solve simultaneous equationsSubstitute into $7x + 2y = 5.5$:

$$10.5 + 2y = 5.5$$

$$y = -2.5$$

FINAL ANSWER

$$x = 1.5, y = -2.5.$$

PAST PAPER QUESTION**Q#: 13****Marks: 6****TOPIC: Differentiation**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

13 The curve **C** has equation $y = 5x^3 - x^2 - 6x + 4$

(a) Find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \dots\dots\dots$$

(2)

There are two points on the curve **C** at which the gradient of the curve is 2

(b) Find the x coordinate of each of these two points.
Show clear algebraic working.

$$\dots\dots\dots$$

(4)

(Total for Question 13 is 6 marks)

PAST PAPER SOLUTION**Q#: 13****Marks: 6****TOPIC: Differentiation**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find the gradient**

$$y = 5x^3 - x^2 - 6x + 4$$

$$\frac{dy}{dx} = 15x^2 - 2x - 6$$

2. Find the gradient

The gradient is 2, so

$$15x^2 - 2x - 6 = 2$$

$$15x^2 - 2x - 8 = 0$$

$$x = \frac{2 \pm \sqrt{(-2)^2 - 4(15)(-8)}}{30}$$

$$x = \frac{2 \pm 22}{30}$$

$$x = \frac{4}{5} \quad \text{or} \quad x = -\frac{2}{3}$$

FINAL ANSWER

$$x = -\frac{2}{3} \text{ and } x = \frac{4}{5}.$$

PAST PAPER QUESTION**Q#: 14****Marks: 3****TOPIC: Expanding Brackets**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

14 Expand and simplify $(4x + 1)(x - 3)(5x + 6)$

(Total for Question 14 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 14****Marks: 3****TOPIC: Expanding Brackets**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Expand brackets**

$$(4x + 1)(x - 3)(5x + 6)$$

$$(x - 3)(5x + 6) = 5x^2 - 9x - 18$$

$$(4x + 1)(5x^2 - 9x - 18) = 20x^3 - 31x^2 - 81x - 18$$

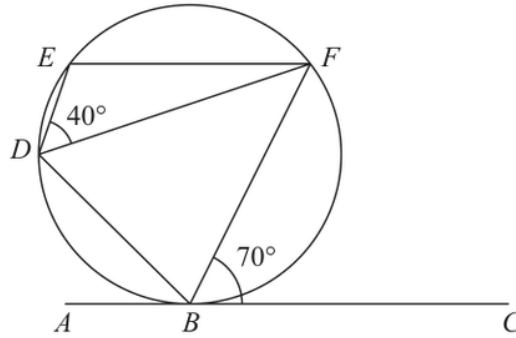
FINAL ANSWER

$$20x^3 - 31x^2 - 81x - 18$$

EA

PAST PAPER QUESTION**Q#: 15****Marks: 4****TOPIC: Circle Theorems**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

15Diagram **NOT**
accurately drawn

B, D, E and F are points on a circle.
 ABC is the tangent to the circle at B .

Angle $EDF = 40^\circ$

Angle $FBC = 70^\circ$

Prove that the tangent ABC is parallel to EF .
 Give a reason for each stage of your working.

(Total for Question 15 is 4 marks)

PAST PAPER SOLUTION**Q#: 15****Marks: 4****TOPIC: Circle Theorems**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Proof**

By the alternate segment theorem, the angle between tangent BC and chord BF equals the angle in the opposite segment:

$$\angle BEF = 70^\circ$$

Given

$$\angle EDF = 40^\circ$$

Angles EDF and EBF stand on the same chord EF , so

$$\angle EBF = 40^\circ$$

In triangle BEF ,

$$\angle EFB = 180^\circ - 70^\circ - 40^\circ = 70^\circ$$

So

$$\angle EFB = \angle FBC = 70^\circ$$

These are alternate interior angles, so

$$EF \parallel ABC$$

FINAL ANSWER

The tangent ABC is parallel to EF .

PAST PAPER QUESTION**Q#: 16****Marks: 9****TOPIC: Functions**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

16 The functions f and g are defined as

$$f: x \mapsto 5x - 7$$

$$g: x \mapsto \frac{5x}{x + 4}$$

(a) Write down the value of x that must be excluded from any domain of g
(1)(b) Find $gf(2.6)$
(2)(c) Solve $fg(x) = 2$ $x =$
(3)(d) Express the inverse function g^{-1} in the form $g^{-1}: x \mapsto \dots$ $g^{-1}: x \mapsto$
(3)**(Total for Question 16 is 9 marks)**

PAST PAPER SOLUTION**Q#: 16****Marks: 9****TOPIC: Functions**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Evaluate fraction**

$$f(x) = 5x - 7, \quad g(x) = \frac{5x}{x+4}$$

2. (a) The denominator of cannot(a) The denominator of g cannot be zero, so

$$x = -4$$

3. (b)

(b)

$$f(2.6) = 5(2.6) - 7 = 6$$

$$gf(2.6) = g(6) = \frac{5(6)}{6+4} = 3$$

4. (c)

(c)

$$fg(x) = 2$$

$$f(g(x)) = 2$$

$$5g(x) - 7 = 2$$

$$g(x) = \frac{9}{5}$$

$$\frac{5x}{x+4} = \frac{9}{5}$$

$$25x = 9x + 36$$

$$16x = 36$$

$$x = \frac{9}{4}$$

5. (d) Let

(d) Let

$$y = \frac{5x}{x+4}$$

$$y(x+4) = 5x$$

$$yx + 4y = 5x$$

$$x(y-5) = -4y$$

$$x = \frac{4y}{5-y}$$

6. Evaluate fraction

So

$$g^{-1}(x) = \frac{4x}{5-x}$$

FINAL ANSWER

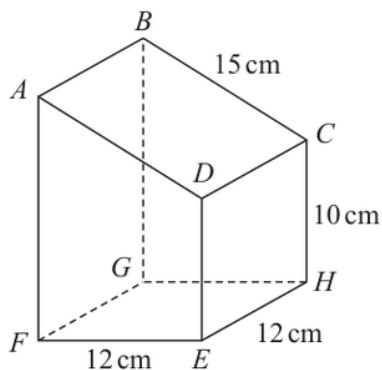
$$-4, 3, x = \frac{9}{4}, \text{ and } g^{-1}(x) = \frac{4x}{5-x}.$$

EA

PAST PAPER QUESTION**Q#: 17** **Marks: 6****TOPIC: 3D Pythagoras & Trigonometry**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

- 17 The diagram shows a prism $ABCDEFGH$ with a horizontal base.

Diagram **NOT**
accurately drawn

The base of the prism, $EFGH$, is a square of side 12 cm.

Trapezium $ADEF$ is a cross section of the prism where AF and DE are vertical edges.

$$DE = CH = 10 \text{ cm}$$

$$AD = BC = 15 \text{ cm}$$

- (a) Work out the size of the angle between CF and the base $EFGH$.
Give your answer correct to one decimal place.

.....
(3)

- (b) Work out the length of BE .
Give your answer correct to one decimal place.

..... cm
(3)

(Total for Question 17 is 6 marks)

PAST PAPER SOLUTION**Q#: 17****Marks: 6****TOPIC: 3D Pythagoras & Trigonometry**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**(a) C is vertically above H , so the vertical height from F to C is 10 cm.**2. The projection of on the**The projection of CF on the base is FH :

$$FH = 12\sqrt{2}$$

$$\tan \theta = \frac{10}{12\sqrt{2}}$$

$$\theta = 30.508\dots^\circ$$

3. Answer**FINAL ANSWER**

30.5°.

4. (b) In the trapezium cross(b) In the trapezium cross-section, $AD = 15$ and the horizontal distance is 12, so the difference in vertical heights is

$$\sqrt{15^2 - 12^2} = 9$$

5. Use trigonometrySince $DE = 10$, the height $BG = 19$ cm.**6. The projection of on the**The projection of BE on the base is $GE = 12\sqrt{2}$, so

$$BE = \sqrt{19^2 + (12\sqrt{2})^2} = 25.475\dots$$

FINAL ANSWER

25.5 cm.

PAST PAPER QUESTION

Q#: 18

Marks: 4

TOPIC: Sequences

May-Nov 2020 P1HR - MayNov 2020 | P1HR

- 18** In an arithmetic series, the 6th term is 39
In the same arithmetic series, the 19th term is 7.8

Work out the sum of the first 25 terms of the arithmetic series.

(Total for Question 18 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 18****Marks: 4****TOPIC: Sequences**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Let the first term be**Let the first term be a and the common difference be d .**2. The 6th term is 39**

The 6th term is 39:

$$a + 5d = 39$$

3. Find the nth term

The 19th term is 7.8:

$$a + 18d = 7.8$$

4. Subtract

Subtract:

$$13d = -31.2$$

$$d = -2.4$$

5. Find the nth term

Then

$$a + 5(-2.4) = 39$$

$$a = 51$$

6. The sum of the first

The sum of the first 25 terms is

$$S_{25} = \frac{25}{2} (2a + 24d)$$

$$S_{25} = \frac{25}{2} (102 - 57.6) = 555$$

FINAL ANSWER

555.

PAST PAPER QUESTION**Q#: 19****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

- 19** The diagram shows rectangle $ABCD$ with rectangle $EFGH$ cut out to form the shaded region.

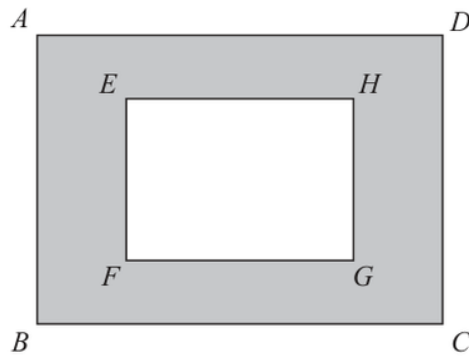


Diagram **NOT**
accurately drawn

$AD = 8.3$ cm correct to one decimal place

$DC = 7.2$ cm correct to one decimal place

$EH = 6.2$ cm correct to one decimal place

$HG = 5.3$ cm correct to one decimal place

Work out the upper bound of the area of the shaded region.
Show your working clearly.

..... cm^2

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION**Q#: 19****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find upper bound**

For the upper bound of the shaded area, use the upper bound of the outer rectangle and subtract the lower bound of the cut-out rectangle.

2. Outer rectangle

Outer rectangle:

$$AD < 8.35, \quad DC < 7.25$$

3. Cut out rectangle

Cut-out rectangle:

$$EH \geq 6.15, \quad HG \geq 5.25$$

$$\text{upper shaded area} = 8.35(7.25) - 6.15(5.25)$$

$$= 60.5375 - 32.2875 = 28.25$$

FINAL ANSWER

$$28.25 \text{ cm}^2$$

PAST PAPER QUESTION**Q#: 20****Marks: 2****TOPIC: Transformations of Graphs**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

20 A curve has equation $y = f(x)$

There is only one maximum point on the curve.

The coordinates of this maximum point are $(-3, 4)$

Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x) - 6$

(.....,)

(ii) $y = f(2x)$

(.....,)

(Total for Question 20 is 2 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 2****TOPIC: Transformations of Graphs**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. The maximum point of is**The maximum point of $y = f(x)$ is $(-3, 4)$.**2. Calculate value**

For

$$y = f(x) - 6$$

3. the graph moves 6 units

the graph moves 6 units down:

$$(-3, 4) \rightarrow (-3, -2)$$

4. Calculate value

For

$$y = f(2x)$$

5. the coordinate is divided bythe x -coordinate is divided by 2:

$$(-3, 4) \rightarrow \left(-\frac{3}{2}, 4\right)$$

FINAL ANSWER $(-3, -2)$ and $\left(-\frac{3}{2}, 4\right)$

PAST PAPER QUESTION**Q#: 20****Marks: 2****TOPIC: Transformations of Graphs**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

20 A curve has equation $y = f(x)$

There is only one maximum point on the curve.

The coordinates of this maximum point are $(-3, 4)$

Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x) - 6$

(..... ,)

(ii) $y = f(2x)$

(..... ,)

(Total for Question 20 is 2 marks)

EA

PAST PAPER SOLUTION**Q#: 20****Marks: 2****TOPIC: Transformations of Graphs**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. The maximum point of is**The maximum point of $y = f(x)$ is $(-3, 4)$.**2. Calculate value**

For

$$y = f(x) - 6$$

3. the graph moves 6 units

the graph moves 6 units down:

$$(-3, 4) \rightarrow (-3, -2)$$

4. Calculate value

For

$$y = f(2x)$$

5. the coordinate is divided bythe x -coordinate is divided by 2:

$$(-3, 4) \rightarrow \left(-\frac{3}{2}, 4\right)$$

FINAL ANSWER $(-3, -2)$ and $\left(-\frac{3}{2}, 4\right)$

PAST PAPER QUESTION**Q#: 21****Marks: 5****TOPIC: Algebraic Roots & Indices**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

21 Given that $M = \frac{18^{4n} \times 2^{3(n^2-6n)} \times 3^{2(1-4n)}}{12^2}$

find the values of n for which $M = 2$

(Total for Question 21 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 21****Marks: 5****TOPIC: Algebraic Roots & Indices**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use index laws**

$$18^{4n} = (2 \times 3^2)^{4n} = 2^{4n} 3^{8n}$$

$$12^2 = (2^2 \times 3)^2 = 2^4 3^2$$

2. Use index laws

So

$$M = \frac{2^{4n} 3^{8n} \times 2^{3(n^2-6n)} \times 3^{2(1-4n)}}{2^4 3^2}$$

3. Combine powers of 2

Combine powers of 2:

$$2^{4n+3n^2-18n-4} = 2^{3n^2-14n-4}$$

4. Combine powers of 3

Combine powers of 3:

$$3^{8n+2-8n-2} = 3^0 = 1$$

5. Use index laws

So

$$M = 2^{3n^2-14n-4}$$

6. Use index lawsGiven $M = 2$,

$$3n^2 - 14n - 4 = 1$$

$$3n^2 - 14n - 5 = 0$$

$$(3n + 1)(n - 5) = 0$$

FINAL ANSWER

$$n = -\frac{1}{3} \text{ or } n = 5$$

PAST PAPER QUESTION**Q#: 21****Marks: 5****TOPIC: Algebraic Roots & Indices**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

21 Given that $M = \frac{18^{4n} \times 2^{3(n^2-6n)} \times 3^{2(1-4n)}}{12^2}$

find the values of n for which $M = 2$

(Total for Question 21 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 21****Marks: 5****TOPIC: Algebraic Roots & Indices**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Evaluate fraction**

$$M = \frac{18^{4n} \times 2^{3(n^2-6n)} \times 3^{2(1-4n)}}{12^2}$$

2. Use index laws

Write everything in powers of 2 and 3:

$$18^{4n} = (2 \cdot 3^2)^{4n} = 2^{4n} 3^{8n}$$

$$2^{3(n^2-6n)} = 2^{3n^2-18n}$$

$$3^{2(1-4n)} = 3^{2-8n}$$

$$12^2 = (2^2 \cdot 3)^2 = 2^4 3^2$$

3. Calculate value

So

$$M = 2^{4n+3n^2-18n-4} \times 3^{8n+2-8n-2}$$

$$M = 2^{3n^2-14n-4}$$

4. Calculate valueGiven $M = 2$,

$$2^{3n^2-14n-4} = 2^1$$

$$3n^2 - 14n - 4 = 1$$

$$3n^2 - 14n - 5 = 0$$

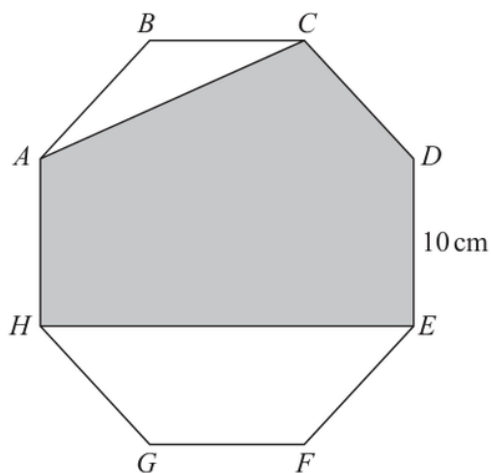
$$(3n + 1)(n - 5) = 0$$

FINAL ANSWER

$$n = 5 \text{ or } n = -\frac{1}{3}$$

PAST PAPER QUESTION**Q#: 22****Marks: 6****TOPIC: Area & Perimeter**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

22 The diagram shows a regular octagon $ABCDEFGH$.Diagram **NOT**
accurately drawn

Each side of the octagon has length 10 cm.

Find the area of the shaded region $ACDEH$.Give your answer correct to the nearest cm^2 cm^2 **(Total for Question 22 is 6 marks)**

PAST PAPER SOLUTION**Q#: 22****Marks: 6****TOPIC: Area & Perimeter**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Each side of the regular**

Each side of the regular octagon is 10 cm.

2. Calculate areaFor a regular octagon with side length s ,

$$\text{area} = 2(1 + \sqrt{2})s^2$$

3. Calculate area

So the area of the whole octagon is

$$2(1 + \sqrt{2})(10)^2 = 200 + 200\sqrt{2}$$

4. The unshaded triangle hasThe unshaded triangle ABC has

$$AB = 10, \quad BC = 10, \quad \angle ABC = 135^\circ$$

5. Use trigonometry

So

$$\text{area of triangle } ABC = \frac{1}{2}(10)(10) \sin 135^\circ = 25\sqrt{2}$$

6. Find the gradientAt the bottom, the unshaded trapezium $HEFG$ has parallel sides

$$HE = 10 + 10\sqrt{2}, \quad GF = 10$$

7. Simplify surd

and height

$$5\sqrt{2}$$

8. Calculate area

So

$$\begin{aligned} \text{area of trapezium } HEFG &= \frac{1}{2}(10 + 10 + 10\sqrt{2})(5\sqrt{2}) \\ &= 50 + 50\sqrt{2} \end{aligned}$$

9. The shaded area is

The shaded area is

$$(200 + 200\sqrt{2}) - 25\sqrt{2} - (50 + 50\sqrt{2})$$

$$= 150 + 125\sqrt{2}$$

$$= 326.7766 \dots$$

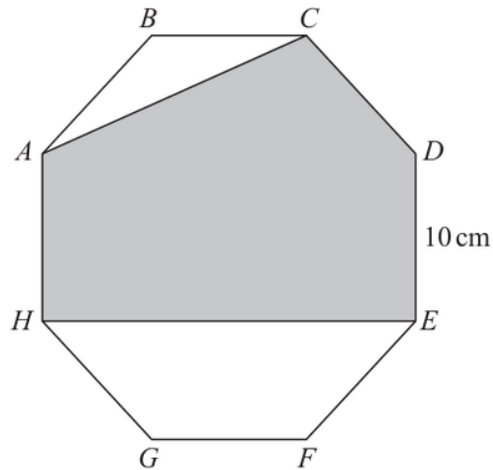
FINAL ANSWER

$$327 \text{ cm}^2$$

EA

PAST PAPER QUESTION**Q#: 22****Marks: 6****TOPIC: Area & Perimeter**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

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PAST PAPER SOLUTION**Q#: 22****Marks: 6****TOPIC: Area & Perimeter**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

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FINAL ANSWER

$$327 \text{ cm}^2$$

EA

PAST PAPER QUESTION

Q#: 23

Marks: 4

TOPIC: **Probability Toolkit**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

23 In a bag, there are only

3 blue beads
4 white beads
and x orange beads.

Jean is going to take at random two beads from the bag.

The probability that Jean will take two beads of the same colour is $\frac{3}{8}$

Find the total number of beads in the bag.
Show clear algebraic working.

(Total for Question 23 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 23****Marks: 4****TOPIC: Probability Toolkit**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. The total number of beads**

The total number of beads is

$$x + 7$$

2. The probability of choosing two

The probability of choosing two beads of the same colour is

$$\frac{\binom{3}{2} + \binom{4}{2} + \binom{x}{2}}{\binom{x+7}{2}}$$

3. Evaluate fractionThis is equal to $\frac{3}{8}$, so

$$\frac{3 + 6 + \frac{x(x-1)}{2}}{\frac{(x+7)(x+6)}{2}} = \frac{3}{8}$$

$$\frac{x^2 - x + 18}{(x+7)(x+6)} = \frac{3}{8}$$

$$8(x^2 - x + 18) = 3(x+7)(x+6)$$

$$8x^2 - 8x + 144 = 3x^2 + 39x + 126$$

$$5x^2 - 47x + 18 = 0$$

$$(x-9)(5x-2) = 0$$

4. Calculate valueSince x must be a whole number,

$$x = 9$$

5. Calculate value

So the total number of beads is

$$9 + 7 = 16$$

FINAL ANSWER

16 beads

PAST PAPER QUESTION

Q#: 23

Marks: 4

TOPIC: **Probability Toolkit**

May-Nov 2020 P1HR - MayNov 2020 | P1HR

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Show clear algebraic working.

(Total for Question 23 is 4 marks)

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May-Nov 2020 P1HR - MayNov 2020 | P1HR

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FINAL ANSWER

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